AN IMMERSION OF ALGEBRAIC STACKS IS UNRAMIFIED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The trivial implications among properties of morphisms of algebraic stacks, in particular that a representable morphism is DM. Invoked inside the counted proof (source lines 8327-8328). Gives that the immersion is DM.
- The statements for schemes that an open immersion and a closed immersion are unramified. Invoked inside the counted proof (source lines 8331-8333) as cross-chapter pointers. Provides unramifiedness of the induced morphism of locally closed subschemes.

Source reference. Original lemma lemma-trivial-implications: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4773–4773. Navigation pointer only: the counted text refers to the Stacks item lemma-trivial-implications, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference morphisms-lemma-closed-immersion-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8333–8333. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-closed-immersion-unramified, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference morphisms-lemma-open-immersion-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8332–8332. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-open-immersion-unramified, which belongs to another chapter

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Lemma 0.1. *An immersion is unramified.*

Proof. Let $j : \mathcal{Z} \to \mathcal{X}$ be an immersion of algebraic stacks. Since j is representable, it is DM by Lemma lemma-trivial-implications. On the other hand, if $X \to \mathcal{X}$ is a smooth and surjective morphism where X is a scheme, then $Z = \mathcal{Z} \times_{\mathcal{X}} X$ is a locally closed subscheme of X. Hence $Z \to X$ is unramified (Morphisms, Lemmas morphisms-lemma-open-immersion-unramified and morphisms-lemma-closed-immersion-unramified) and we conclude that j is unramified from the definition. $\square$